

CarDD: Detecting Car Damage with Modern YOLO Detectors

Dataset — CarDD

A YOLO-format detection dataset of **4,000 images** and **8,740 annotated damage boxes** across **6 classes**, trained at **1024×1024** resolution.

Split sizes	
Split	Images
Train	2,816
Validation	810
Test	374
Total	4,000

Boxes per class	
Class	Boxes
scratch	3,595
dent	2,543
crack	898
lamp broken	704
glass shatter	681
tire flat	319
Total	8,740

Challenge: strong class imbalance (scratch is 11× tire flat) and many small, low-contrast damage regions — motivating the high-resolution P2 head.

Models & Training

Both detectors share an identical data pipeline and schedule (Ultralytics).

	YOLO11-L	YOLOv8-L-P2
Init weights	yolo11l.pt	yolov8l-p2
Extra P2 head	no	yes
Image size	1024	1024
Max epochs	150	150
Early stop @	106	46
Patience	30	30
Optimizer	auto	auto
Initial LR	0.01	0.01

Pipeline: EDA & class-balance → enhancement + augmentation → training (early stop, patience 30) → evaluation on held-out **test** split → head-to-head comparison.

Qualitative Results — Test Predictions

Predictions on identical test images (conf=0.25)

Predicted damage boxes with class & confidence on unseen test images.

Training Curves & PR Analysis

Training results.png

Left: loss & mAP convergence. Right: per-class Precision–Recall curves on the test split.

Results — Overall (Test Split)

Model	mAP@50	@50-95	@75	P	R
YOLO11-L	0.725	0.557	0.589	0.795	0.668
YOLOv8-L-P2	0.721	0.541	0.571	0.748	0.701

YOLO11-L leads on mAP & precision; YOLOv8-L-P2 gives slightly higher recall.

Results — Per-Class AP@50-95

Class	YOLO11-L	YOLOv8-L-P2
glass shatter	0.860	0.860
tire flat	0.859	0.839
lamp broken	0.716	0.689
dent	0.344	0.323
scratch	0.309	0.281
crack	0.253	0.256

Rigid, high-contrast damage scores well; thin **crack**/scratch stay hardest.

Conclusion

- **YOLO11-L is the best overall model:** mAP@50-95 **0.557** vs. 0.541, precision 0.795.
- Wins **5 / 6** classes on AP@50-95; YOLOv8-L-P2 edges it only on **crack** (+0.003).
- The extra P2 head did not overturn YOLO11-L on small damage, yet plateaued far sooner (early stop @46).
- **Future work:** oversampling/augmentation for crack & tire flat, attention for fine cracks, and test-time augmentation.